

Cost-Optimized Diploma Verification on Ethereum Using Per-Record and Merkle-Batch Storage Architectures

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Abstract—Fake academic diplomas are a growing problem worldwide, yet checking whether a diploma is real still requires slow, manual contact with the issuing university. Blockchain registries can solve this by letting anyone verify a credential on-chain—but on Ethereum, storing data is expensive, so the way diplomas are represented in a smart contract matters greatly.

This paper examines these costs using *UniVerify*, a diploma-verification system built on Ethereum. Two designs are compared: *per-record storage*, where each diploma takes one contract slot, and *Merkle-batch issuance*, where a whole graduation class is covered by a single cryptographic root. To separate the effect of the architecture from code-level choices, four Solidity contract variants are tested with batch sizes from 1 to 4,096 diplomas.

The difference is substantial. Merkle batching cuts the per-diploma issuance cost by 98.9 % for a class of 100 students and by three orders of magnitude at 1,000 students—from \$4,820 down to \$5.09 at current Ethereum prices. Low-level EVM optimizations save another 6.6 % on verification gas, but they are far less important than the choice of architecture. A closed-form model is also derived that decomposes the optimization savings into a per-step and a constant term and matches measured gas values with errors below 0.2 %.

The main conclusion is that for universities that issue diplomas in batches, the data structure used on-chain is the most important cost factor—not low-level code tuning—and Merkle-batch registries are the practical default for blockchain-based academic credentials.

Index Terms—Ethereum, Solidity, gas optimization, Merkle tree, smart contracts, diploma verification

I. INTRODUCTION

The market for fake academic credentials is estimated at over one billion dollars per year [1], but verifying whether a diploma is genuine still depends on contacting the issuing university—a process that is slow and hard to scale across borders. Blockchain-based registries offer a different approach: once a diploma commitment is saved on-chain, anyone can check it independently, without trusting a single database.

The main obstacle is cost. On Ethereum, every permanent storage write costs about 20,000 gas. If each diploma is stored one by one, a university issuing hundreds or thousands of diplomas per graduation cycle faces very high fees. The question is therefore not *whether blockchain* can be used for credentials, but *how diplomas should be stored on-chain so that costs stay reasonable*.

This paper addresses that question using *UniVerify*, a prototype diploma-verification system consisting of a Solidity smart contract and an off-chain TypeScript SDK. Universities act as issuers who publish diploma commitments on-chain, while verifiers check a diploma by providing the document hash and any necessary cryptographic proofs. Two fundamental approaches are compared: (1) **Per-record storage**, where each diploma is saved in its own mapping slot and verified in constant time; and (2) **Merkle batch issuance**, where one root covers an entire graduation class, and diplomas are verified via off-chain proofs.

The main contribution of this work is a practical cost evaluation of real, deployable smart contracts. Specifically, this paper provides:

- 1) A cost comparison of packed per-record storage and Merkle-batch issuance across deployment, issuance, verification, and revocation.
- 2) A four-way benchmark that separates architectural, library, and EVM-level effects.
- 3) A closed-form linear gas-savings model validated at six proof depths with errors below 0.2%.
- 4) An economic analysis translating gas savings into real dollar costs for institutions, including market sensitivity analysis.

II. RELATED WORK

Grech and Camilleri [2] identified credential verification as one of the main uses of blockchain in education. Lizcano et al. [4] proposed a decentralized credential system using IPFS and Ethereum but did not analyze gas costs in detail. Alketbi et al. [3] surveyed blockchain applications in government services, including education records. The W3C Verifiable Credentials standard [10] defines a data model for tamper-proof claims but does not address cost. Arndt et al. [11] measured on-chain credential storage on Ethereum L2 networks but did not compare Merkle batching with per-record storage.

Merkle trees [5] are the standard way to prove that an element belongs to a set. OpenZeppelin's *MerkleProof* [9] is the most widely used Solidity library for this purpose. Other approaches—sparse Merkle trees, Verkle trees [12], RSA accumulators—are not yet practical for typical EVM contracts.

Ethereum execution costs are defined by the EVM gas schedule [8], and EIP-2929 increased costs for first-time storage

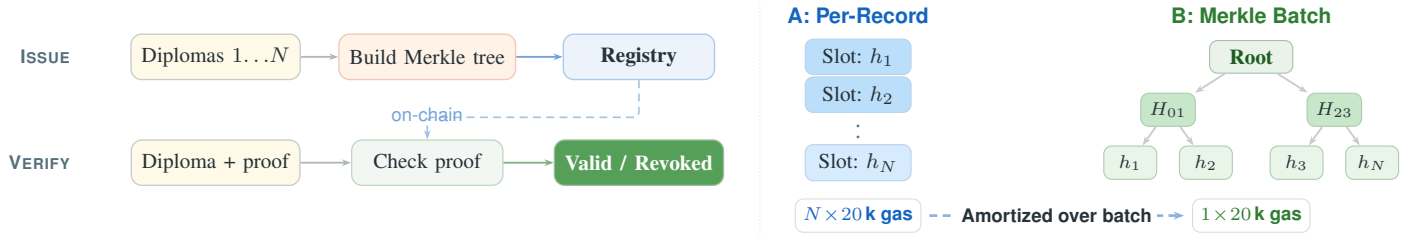


Fig. 1: UniVerify system architecture. *Left*: issuance and verification workflow (off-chain view calls cost 0 gas). *Right*: on-chain cost comparison of the two issuance models.

TABLE I: Contract Variants and Key Design Choices

Label	Arch.	Key design choice
Packed	Per-rec.	Packed slot: issuer (160b) + revoked (1b) in one <code>uint256</code>
Baseline	Merkle	Hand-written proof loop; leaf via <code>abi.encodePacked</code>
OpenZeppelin	Merkle	OZ <code>MerkleProof.verifyCalldata</code>
Optimized	Merkle	Four EVM-level optimizations (T1–T4) on the hot path

access [6]. Most earlier work on blockchain credentials focuses on architecture, privacy, or governance. This paper takes a different approach: it measures how specific Solidity design choices affect the gas cost of issuing and verifying diplomas in an implemented contract.

III. SYSTEM DESIGN AND METHODOLOGY

A. Architectures and Contract Variants

As illustrated in Fig. 1, two fundamental architectures were evaluated. In **Architecture A (Per-Record Storage)**, each diploma is stored in a mapping (`bytes32` $\Rightarrow$ `uint256`). The issuer address (160 bits) and a revocation flag (1 bit) are packed into one 256-bit slot. Issuing a diploma requires one `SSTORE`, and verifying it requires one `SLOAD` plus bit masking, yielding a constant read cost. In **Architecture B (Merkle Batch)**, all diplomas in a graduation class form a Merkle tree off-chain. Only the 32-byte root is written on-chain, covering the whole batch in a single storage write. Verification requires the caller to provide $d = \lceil \log_2 N \rceil$ sibling hashes. Each leaf utilizes domain separation to prevent cross-chain replay attacks.

To isolate the architectural effects from implementation choices, four Solidity contract variants were built (Table I). Packed implements the per-record storage baseline. Baseline uses a hand-written Merkle proof loop with `abi.encodePacked`. OpenZeppelin utilizes the widely adopted OpenZeppelin `MerkleProof` library. Finally, Optimized introduces targeted EVM-level optimizations on the verification path to test the limits of gas reduction.

B. Optimization Techniques and Experimental Setup

The Optimized variant incorporates four low-level techniques to minimize verification costs. **T1 (Unchecked loop)**:

`unchecked{++i;}` removes arithmetic overflow checks during proof iteration. **T2 (Scratch-space hashing)**: Inputs are placed directly in EVM scratch memory and hashed in place, avoiding memory allocation overhead. **T3 (Validation splitting)**: Public functions validate inputs, while internal paths skip redundant checks. **T4 (Assembly preimage)**: The 112-byte leaf preimage is packed using `mstore/shl` instead of Solidity’s native encoding.

All measurements were performed using Foundry [7] with Solidity 0.8.20 and the optimizer set to 200 runs. Proof depths $d \in \{0, 1, 4, 8, 10, 12\}$ were evaluated, corresponding to batch sizes from 1 to 4,096 diplomas. Function-level gas consumption was extracted directly from Forge test reports.

IV. RESULTS

A. Gas Consumption Analysis

Table II compares the per-diploma issuance cost of the two architectures. Publishing a Merkle batch root costs 50,939 gas no matter how many diplomas are in the batch; the amortized cost per diploma is therefore $50,939/N$ (Fig. 2). In contrast, per-record issuance costs a fixed 48,200 gas for every single diploma. While batching is slightly more expensive for a single credential, amortization becomes highly advantageous starting from $N \geq 2$. At $N=100$, Merkle issuance is 98.9% cheaper per diploma.

Merkle batching inherently shifts a portion of the cost from issuance to verification. Table III highlights this trade-off. Per-record verification costs only 2,720 gas—just one storage read plus bit masking. The Merkle variants incur higher verification costs due to proof processing. Among them, Optimized achieves the lowest average gas, saving 6.7% versus Baseline and 3.1% versus OpenZeppelin.

Regarding deployment, Packed requires approximately one-third the gas of the Merkle variants due to its simpler logic (Table IV). However, deployment is a one-time cost, whereas

TABLE II: Per-Diploma Issuance Cost

N	Merkle (gas)	Per-record (gas)	Ratio
1	50,939	48,200	$0.95 \times$
10	5,094	48,200	$9.5 \times$
50	1,019	48,200	$47 \times$
100	509	48,200	$94.7 \times$
1,000	51	48,200	$945 \times$

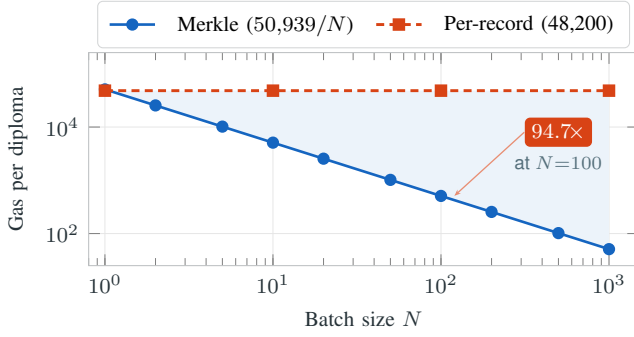


Fig. 2: Per-diploma issuance cost vs. batch size. The shaded area shows how much Merkle batching saves over per-record storage.

TABLE III: Verification Gas across Variants

Variant	Min	Avg	Max	vs. Pkd (%)
Packed	2,720 (constant)			—
Optimized	10,417	11,104	12,103	4.1× (−6.7%)
OpenZeppelin	10,560	11,532	12,936	4.2× (−3.1%)
Baseline	10,561	11,896	13,841	4.4× (ref.)

issuance is a recurring expense each graduation cycle, making the latter far more impactful over the contract’s lifespan.

B. Economic Impact and Analytical Model

To contextualize these gas metrics, we apply an economic projection assuming a gas price of $g=40$ gwei and $p=\$2,500/\text{ETH}$. This yields an issuance cost of \$4.82 per diploma for Packed and \$5.09 for one total Merkle batch root. At an institutional scale, publishing 1,000 diplomas costs \$4,820 with per-record storage, compared to just \$5.09 with batching.

To fully capture this architectural advantage against cryptocurrency volatility, Table V presents a sensitivity matrix of batch issuance costs under varying network states. Even under high network congestion (80 gwei, \$4,000/ETH), the 1,000-diploma batch cost remains under \$17, solidifying the model’s robustness.

TABLE IV: Deployment and State-Changing Costs (gas)

Variant	Deploy	Revoke (avg)	Issue
Packed	393,596	29,302	48,200
Optimized	1,018,273	58,303	50,939
OpenZeppelin	1,036,010	56,871	50,939
Baseline	1,142,020	59,610	51,012

TABLE V: Cost of Issuing 1,000 Diplomas

Gas Price & ETH	Merkle (\$)	Per-record (\$)	Savings
Low (15 gwei, \$2k)	1.53	1,446.00	99.89%
Base (40 gwei, \$2.5k)	5.09	4,820.00	99.89%
High (80 gwei, \$4k)	16.30	15,424.00	99.89%

TABLE VI: Closed-Form Model Validation

d	Leaves	Model Δ	Measured	Residual
0	1	144	144	0
1	2	343	343	0
4	16	940	940	0
8	256	1,736	1,738	2 (0.12%)
10	1,024	2,134	2,137	3 (0.14%)
12	4,096	2,532	2,536	4 (0.16%)

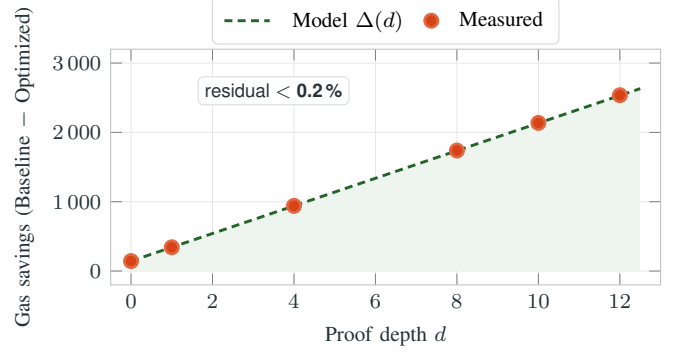


Fig. 3: Closed-form model (1) compared with measured optimization savings.

The gas savings achieved by Optimized over Baseline can be mathematically decomposed into a linear model. Let d be the proof depth:

$$\Delta(d) = \underbrace{d \cdot 199}_{\text{T1+T2: per proof step}} + \underbrace{144}_{\text{T3+T4: constant}} \quad (1)$$

The linear coefficient (199 gas) corresponds precisely to the per-iteration savings from bypassing the ADD opcode overflow check (via `unchecked`) and avoiding memory expansion costs by hashing in the pre-allocated EVM scratch space. The constant term (144 gas) reflects the fixed savings from the unified assembly preimage setup (`mstore/shl`) and reduced jump destinations via validation splitting. Table VI and Fig. 3 confirm that this model matches measured gas values with exceptional accuracy, showing a residual error below 0.2% across all tested depths.

V. DISCUSSION

The core engineering trade-off in credential registries lies between issuance and verification efficiency. The Packed baseline offers the cheapest verification, but its issuance cost grows linearly with the number of diplomas. Merkle batching reverses this dynamic: issuance remains virtually constant at the batch level, but each verification requires an off-chain proof. For academic institutions, this trade-off is highly favorable. Issuance is a paid, on-chain operation borne by the university, while verification is generally treated as a read-only operation executed via local nodes or RPC endpoints, costing zero real ether.

Based on the benchmark results, a clear hierarchy of optimization impact emerges. At the highest level, the architectural

switch to Merkle batching dominates, cutting per-diploma issuance costs by up to 98.9%. At the mid-level, library selection plays a role, with OpenZeppelin’s `MerkleProof` providing a 3.1% reduction in verification gas over an unoptimized baseline. Finally, extreme low-level EVM tuning (Optimized) yields an additional 6.6% reduction in verification gas and a 10.8% reduction in deployment costs. However, these low-level gains cannot compensate for an inefficient storage model, reinforcing that the elimination of per-record state writes is the single most critical optimization.

From a practical standpoint, OpenZeppelin’s standard library is the recommended default for production deployments. It captures the bulk of available gas savings while maintaining code readability and ease of auditing. Hand-tuned assembly is only justified under extreme throughput requirements where verification costs directly translate to execution constraints. Conversely, for highly localized, small-scale registries updating a handful of records sporadically, a simple packed per-record storage approach remains a viable and robust option.

This study possesses certain limitations. Measurements were strictly confined to Ethereum Layer 1 gas dynamics and isolated to warm EVM storage access (100 gas per `SLOAD`) using Foundry. EIP-2929 cold-access surcharges on mainnet impose an initial 2,100 gas penalty per slot read. While this would increase absolute verification costs for all variants, it does not alter their relative ranking, as both architectures require at least one cold storage read to fetch either the credential root or the per-record flag. Future research should expand this evaluation to Layer 2 networks, calldata-aware designs under EIP-4844, and forthcoming Verkle-tree state architectures.

VI. CONCLUSION

This paper evaluated two deployable architectures for on-chain diploma verification and presented a closed-form model

predicting optimization savings across variable proof depths. The empirical evidence demonstrates that architectural design supersedes code-level optimization in reducing Ethereum execution costs. For universities issuing credentials in cohorts, Merkle batching decreases the per-diploma issuance cost by up to three orders of magnitude compared to traditional per-record storage mappings. As academic registries increasingly migrate to blockchain infrastructure, these findings establish Merkle-based issuance combined with standard cryptographic libraries as the most economically viable framework, ensuring that decentralized verification remains sustainable at scale.

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